# In-Class Work Session

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**Goal:** Leave with a concrete, feasible project direction that you can develop into your project proposal. Complete Sections 1-7 as a group. Limit each response to 1-3 sentences.

- Form a brainstorming group in one of these areas: **Computer Vision** · **NLP / LLMs** · **Sciences** (bio, chem, physics, climate) · **Networks / Graphs** · **Signals / Audio**.
- Want a different area? Email cpsc4710_fall26_staff@googlegroups.com before the session with a one-line reason.

**Team members:** _____

**Application area:** _____

**Submission Instructions:** Submit your answers to the mandatory questions on Canvas by **Wednesday October 14, 2026 at 11:59pm ET**.

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## (Mandatory): Brainstorming Questions

Answer the questions below in 1-3 sentences. Use the optional guiding questions and resources on the next few pages if your group gets stuck.

1. **What task will you study? What dataset(s) and baseline model will you start with? Identify one primary trust axis and one secondary trust axis to address.**
    - What publicly available dataset(s) can you use for this task? What existing models provide reasonable baselines? Be specific enough so another group could understand your experimental setup.
    - Consider trust axes like explainability, robustness, privacy, fairness, etc.
2. **What is the proposed method you will implement, adapt, or change to address the problem?**
    - This can be an intervention at the data, training, or inference stage.
    - Your method does not need to be finalized yet.
3. **What evaluation metric will you use, and what stress test or sanity check will you run?**
    - Choose measurements that reflect the trust failure you identified.
    - Stress tests might include corruptions, synthetic confounders, OOD splits, or subgroup analyses.
4. **What are the main data, compute, or timeline constraints? What is the biggest risk and your fallback plan?**
    - Prefer a baseline that is easy to run and plan a smaller pilot before scaling up.

## (Optional): Guiding Questions

These are *guiding questions* you can optionally use to help you answer the questions above. You **do not** need to answer all of them. Revisit these questions in your project proposal.

1. **Task, Dataset, Baselines, and Trust Axes:**
    a. Why is trust necessary for your application area? Why might users not trust your baseline model?
    b. Who are the relevant stakeholders (e.g., end users, clinicians, etc.)?
2. **Proposed Method:**
    a. What existing methods have been proposed for your trust objective? What is the core idea behind those methods?
    b. If you plan to modify an existing method, what's the justification for your proposed modification?
3. **Evaluation Metrics and Stress Tests:**
    a. Are there any limitations with your chosen metrics?
4. **Feasibility, Risks, and Fallbacks:**
    a. How much preprocessing will the data require?
    b. What is the minimum viable experiment you could run that would still answer your research question?

## (Optional): Resources

An excellent starting point would be the reading materials (surveys, papers) mentioned in slides. You can also explore the following sites for example implementations:

- Papers With Code
- Search "github awesome <your topic>" for curated resources (e.g., "github awesome LLM interpretability")

These tutorials may also be useful for inspiration or scoping:

- **Explainability -** Image and Text Classification with LIME
- **Adversarial Robustness -** Self-Study on Adversarial Robustness
- **Privacy -** PyTorch: From Centralized to Federated Learning
- **Fairness -** Learning a Fair Loss Function in PyTorch
- **Efficiency -** Post-Training Quantization
- **Mechanistic Interpretability -** How to Become a Mechanistic Interpretability Researcher